Σ■ Ouro — Combined FC Adapter

Training & Bridge Test Results · 2026-06-22

1 Objective

Train a QLoRA adapter on **Ouro-1.4B-Thinking** that teaches the model both the tool-call trigger decision (when to call vs. when to answer in plain text) and the exact <tool_call>{...}</tool_call> output format, replacing the previous FC adapter that required forced tool_choice for every invocation.

2 Dataset Composition

Source	License	Rows	Purpose
NousResearch/hermes-function-calling-v1	Apache-2.0	8,940	Positive trigger + format
MadeAgents/xlam-irrelevance-7.5k	CC-BY-4.0	6,051	Negative (don't-call) signal
Σ■ coding examples (HumanEval-style)	Internal	243	Coding capability retention
Combined (after BOM-safe merge)	—	15,234	Training pool

3 Training Run

Parameter	Value
Base model	ByteDance/Ouro-1.4B-Thinking
Adapter type	QLoRA (bitsandbytes NF4, bf16)
LoRA rank / alpha	r=32 / α=64 → 30.3 M trainable params (2.07%)
Sequence length	1,536 tokens (seq=4096 overflows VRAM → 39 min/step; seq=1536 → 38 s/step)
Steps / batch	600 steps, batch=1, grad_accum=8 → 4,800 unique examples
LR schedule	2e-4 peak, warmup_ratio=0.03, cosine decay
Optimizer	paged_adamw_8bit, gradient_checkpointing
Wall-clock time	6 h 19 m 56 s (22,736 s)
Final train_loss	0.038 (perplexity 1.04 — near-perfect format learning)
Output adapter path	D:\lantern-train\ouro-sigma0-combined-adapters\final

Loss curve highlights

Step	Loss	LR	Note
10	0.458	1.0e-4	warmup ramp
20	0.128	2.0e-4	peak LR
30	0.060	1.96e-4	rapid descent

40	0.034	1.93e-4	converged
50–600	0.018–0.050	decaying	stable low-loss plateau
600 (final)	0.038	3.4e-7	near-zero LR, done

4 Serving Stack

The adapter is loaded by **ouro_serve.py** (HF generate + PEFT merge_and_unload) behind an Ollama-compatible API on port 11434. The **ouro_anthropic_bridge.py** proxy translates Anthropic `/v1/messages` requests (with `tools[]`) into the training-format prompt and parses `<tool_call>` output back into Anthropic `tool_use` content blocks.

Component	Env var / Flag	Value used
ouro_serve.py	OURO_ADAPTER	D:/lantern-train/ouro-sigma0-combined-adapters/final
	OURO_CANARY	1 (collapse/divergence logits processor)
	OURO_KV_INT8	1 (~halves KV-cache VRAM)
	OURO_ADAPT	1 (enable adapter path)
bridge.py	BRIDGE_PORT	8788
	OURO_OLLAMA_URL	http://127.0.0.1:11434

5 Bridge Test Results

Three-query test (Anthropic /v1/messages via bridge → ouro_serve → Ouro-1.4B):

#	Query	Expected	Got	Result
Q1	What is the weather in Seattle?	tool_call (get_weather)	530-char text — 'I don't have real-time access…	FAIL
Q2	Tell me a funny joke about programmers	plain text	Joke + emoji (no tool call)	PASS
Q3	Write a Python function that reverses a string.	plain text	Starts 'def reverse…' (no tool call)	PASS

Raw `/api/generate` test (exact training-format prompt, bypassing bridge):

Query	Tool Provided	Expected	Result
Search for laptops under $500	search_database	TOOL call	FAIL — refuses ('no products listed')
What is 17 × 342 + 88?	calculate	TOOL call	FAIL — computes directly (wrong: 5906)
Look up profile for user abc-123	get_user_info	TOOL call	FAIL — 'no access to external APIs'
Write a haiku about autumn leaves	get_user_info	plain text	PASS — correct haiku
What is the capital of France?	calculate	plain text	PASS — 'Paris' (minor hallucination)

6 Analysis

Capability	Status	Evidence
Negative discrimination (don't-call)	WORKING	Jokes, coding, geography — all correctly return plain text with no tool_use block
Positive trigger (auto-call without tool_choice)	NOT WORKING	All tool-required queries return text refusals; model never emits <tool_call>
Format learning (loss metric)	STRONG	train_loss 0.038 / perplexity 1.04 — model memorised output format
Pre-training override	DOMINANT	1.4B base has strong 'I don't have real-time access' priors; 2,350 hermes examples (

7 Adapter Comparison

Adapter	Auto-trigger (no tool_choice)	Negative discrimination	Notes
ouro-sigma0-fc-adapters/final (FC-only, ~300 steps)	Partial	POOR	Triggers format reliably with tool_choice=required; calls tools on irrelevant
ouro-sigma0-combined-adapters/final (this run, 600 steps)	NONE	GOOD	Never auto-triggers; correctly refuses irrelevant; still needs tool_choice=re
Target behaviour (future)	FULL	GOOD	2,000+ steps, hermes-heavy mix, or two-phase training

8 Recommended Next Steps

Priority	Action	Expected outcome
HIGH	Re-run training at 2,000–5,000 steps, hermes-only (remove irrelevance negatives for first phase)	Positive trigger activates; loss stays low
HIGH	Add forced-positive few-shot prefix to bridge system prompt (1 example of call + 1 of no-call)	Bridge performance improves with current adapter
MED	Two-phase training: phase 1 hermes-only → phase 2 hermes + negatives (3:1 ratio)	Both positive trigger and negative discrimination
MED	Tune stop-strings: add OURO_NO_STOP=1 on server; verify <tool_call> not being truncated	Eliminates potential stop-string interference
LOW	HumanEval re-run with combined adapter vs FC-only to quantify coding regression (if any)	Baseline comparison for combined vs FC adapter